

UNIT – I INTRODUCTION TO DATABASE SYSTEMS**9L, 8P**

Introduction to Databases - File System Vs Database System - Data Models - Schemas and Instances - DBMS Architecture - Centralized - Client Server - Database Applications - ER Models – ER to Relational Mapping

PRACTICALS:

1. ER Diagram

UNIT – II RELATIONAL MODELS**10L, 24P**

Relational Model - Constraints - Keys - Dependencies - Relational Algebra - Unary, Binary, Set and Extended Relational Algebra operations – SQL- Data Definition - Data Manipulation and Retrieval Queries - Nested Queries – Joins - Views-Cursors -Procedures – Cursors-Procedures - Triggers - Embedded and Dynamic SQL

PRACTICALS:

DDL commands:

1. Creation of tables with appropriate integrity constraints
2. Usage of alter, drop commands

DML commands:

3. Data Insertion, updation, and deletion with tables
4. Data retrieval using
 - Simple SQL
 - Nested Queries
5. Different types of Joins
6. PL/SQL: Functions, Procedures and Triggers

UNIT – III RELATIONAL DATABASE DESIGN**7L, 4P**

Database Design - Functional Dependencies – Normalization - 1 NF - 2 NF - 3 NF - BCNF - Multivalued Dependency (4 NF) - Join Dependency (PJNF)

PRACTICALS:

1. Aggregation operators-Grouping and ordering

UNIT – IV TRANSACTIONS AND RECOVERY**10L, 12P**

Transaction processing concepts - Need for concurrency control and recovery- ACID Properties - Recoverability – Serializability - Concurrency Control - Two phase locking Techniques – Timestamp based protocol – Graph based protocol - Deadlock handling – Log based recovery – Two Phase Commit Protocol

PRACTICALS:

Create View tables

1. DCL Commands: Grant and Revoke
2. TCL Commands: Save point, Commit, Rollback

UNIT – V**QUERY PROCESSING AND ADVANCED DATABASES****9L, 12P**

Indexing and Hashing Techniques - Query Processing and Optimization - Sorting and Joins – Database Tuning – Introduction to Spatial and Temporal Databases – OO Databases – NoSQL

PRACTICALS:

1. Implementation of suitable front end for querying and displaying the results

TOTAL: 45L + 60P = 105 PERIODS**COURSE OUTCOMES:**

Upon completion of the course, the students will be able to

1. Model an application's data requirements using conceptual modeling tools like ER diagrams and design database schemas based on the conceptual model
2. Formulate solutions to a broad range of query problems using relational algebra/ SQL
3. Apply normalization theory to normalize the relations in RDBMS to avoid redundancy and anomalies.
4. Manage concurrent transactions and their consequences, and analyze the use of triggers, functions, and procedures in a realistic database application.
5. Understand database storage structures and access techniques

REFERENCES:

1. Abraham Silberschatz, Henry F. Korth, S. Sudharshan, "Database System Concepts", Seventh Edition, Tata McGraw Hill, 2019.
2. Ramez Elmasri, Shamkant B. Navathe, "Fundamentals of Database Systems", Seventh Edition, Pearson/Addison - Wesley, 2016.
3. C.J. Date, A. Kannan and S. Swamynathan, "An Introduction to Database Systems", Pearson Education, Eighth Edition, 2006.
4. Raghu Ramakrishnan and Johannes Gehrke, "Database Management Systems", Third Edition, McGraw Hill, 2014.
5. Andreas Meier, Michael Kaufmann, "SQL & NoSQL Databases: Models, Languages, Consistency Options and Architectures for Big Data Management", 1st ed. 2019 Edition.
6. Narain Gehani and Melliya Annamalai, "The Database Book: Principles and Practice Using the Oracle Database System", Universities Press, 2012.

CO-PO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	3	3	3	3	1	-	2	-	1	3	3	3	2
CO2	3	3	3	3	3	3	1	-	2	-	1	2	3	3	2
CO3	3	3	3	3	2	3	-	-	3	-	1	3	3	3	2
CO4	3	3	3	3	3	3	-	-	1	-	1	2	3	3	2
CO5	3	3	3	2	3	3	-	-	2	-	1	3	3	3	2

1-low, 2-medium, 3-high, '-'- no correlation